

② expresiones regulares automática

$$q_0 = \varepsilon$$

$$q_1 = q_0 \varepsilon = q_0$$

$$q_2 = q_1 0 = 0$$

$$q_3 = q_0$$

$$q_4 = q_3$$

$$q_5 = q_2 + q_4 = 0 + \varepsilon$$

$$q_6 = q_5$$

$$q_7 = q_6$$

$$q_8 = q_7 1 = (0 + \varepsilon) 1$$

$$q_9 = q_6$$

$$q_{10} = q_9$$

$$q_{11} = q_8 + q_{10} = (0 + \varepsilon) 1 + (0 + \varepsilon) = (0 + \varepsilon)(1 + \varepsilon)$$

$$q_{12} = q_5$$

$$q_{13} = q_{12}$$

$$q_{14} = q_{11} + q_{13} = ((0 + \varepsilon)(1 + \varepsilon)) + (0 + \varepsilon) = (0 + \varepsilon)(1 + \varepsilon)$$

$$q_{15} = q_{14} + q_{16} \varepsilon$$

$$q_{16} = q_{15} 0$$

$$q_{17} = q_{14} + q_{16}$$

$$\text{estado aceptación} = q_{17} = q_{14} \mid q_{16}$$

Lema de Arden

$$r = q + rp$$

$$r = qp^*$$

$$\rightarrow q_0 = \varepsilon$$

$$\rightarrow q_5 = q_2 + q_4 \varepsilon$$

$$q_4 = q_2 \varepsilon^*$$

$$q_1 = \varepsilon + q_1 \emptyset$$

$$\hookrightarrow r = q_1 = qp^* = \varepsilon \emptyset^* = \emptyset^*$$

$$q_1 = \emptyset^*$$

$$q_2 = \emptyset^* 1 + q_2 1$$

$$q_2 = r = qp^* = \emptyset^* 1 1^* = \emptyset^* 1 1^*$$

si concatenamos cualquier e.r con ε nos da la e.r

$$\rightarrow q_{15} = q_{14} + q_{16} \varepsilon = q_{14} + q_{15}^r \varepsilon$$

$$q_{16} = q_{14} \delta^*$$

$$\rightarrow q_{17} = q_{14} + q_{16} \varepsilon \rightarrow q_{16} = q_{14} \delta^*$$

$$q_{17} = q_{14} + q_{14} \delta^* \varepsilon$$

$$= q_{14} \delta^*$$